DevOps Infrastructure – Unit 1 Notes (Part 4)

Image Registries & Distribution, Introduction to Docker

---

1. Image Registries

Definition:

An **image registry** is a centralized storage system used to:

- Store container images
- Manage versions
- Distribute images across systems

Why Registries are Needed

- Share images between developers
- Store application versions
- Enable CI/CD pipelines
- Support deployment across environments

Types of Image Registries

a) Public Registries

- Open to everyone
- Used for sharing common images

Example:

- Docker Hub (default registry for Docker)

---

b) Private Registries

- Restricted access
- Used by organizations

Examples:

- AWS ECR

- Google Container Registry
- Azure Container Registry

Common Operations

Push

Upload image to registry

```
docker push my_app
```

Pull

Download image from registry

```
docker pull my_app
```

---

2. Image Distribution Workflow

Step-by-Step Flow:

1. Developer builds an image
2. Tags the image (versioning)
3. Pushes image to registry
4. Another system pulls the image
5. Container is created from image

Example

```
docker build -t my_app:v1 .
docker push my_app:v1
docker pull my_app:v1
docker run my_app:v1
```

Benefits

- Version control of applications
- Easy rollback to previous versions

- Faster deployment in distributed systems

---

3. Introduction to Docker

Definition:

Docker is an open-source platform used to:

- Build
- Package
- Deploy
- Run containers

It simplifies containerization using easy-to-use tools.

Why Docker is Popular

- Easy to use CLI
- Lightweight containers
- Strong ecosystem
- Widely adopted in industry

Key Features

Portability

- Run anywhere (local, cloud, server)

Isolation

- Apps run independently

Efficiency

- Uses fewer resources than VMs

Scalability

- Easily scale applications

---

4. Basic Docker Workflow

Steps:

1. Write a Dockerfile
2. Build an image
3. Run a container
4. Push image to registry

Example Commands

Build Image:

docker build -t my_app .

Run Container:

docker run my_app

---

5. Real-World DevOps Integration

Docker is used in:

- CI/CD pipelines (Jenkins, GitHub Actions)
- Microservices architecture
- Cloud deployments (AWS, Azure, GCP)

---

6. Summary

- Image registries store and distribute container images
- Public and private registries serve different needs
- Docker simplifies container creation and management
- Core commands: build, push, pull, run
- Essential tool in modern DevOps workflows